Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Use of $P_{\text{max}} = IV$ [at / close to 'knee'] (1) P_{max} = 130 ± 5 [W] (1) $P_{\text{in}} = IA = 600 \times 1.2 = 720$ [W] (1) $\frac{130\ \textbf{ecf}}{720}$ = 0.18 or 18 [%] (1)	1	1 1 1		4	3	
	(d)	(i)	Use of $\frac{\Delta Q}{\Delta t} = KA\frac{\Delta\theta}{\Delta x}$ (1) Answer seen 4900 (1)	1	1		2	1	
		(ii)	Attempt at calculating $\Delta\theta_{\text{for 1 layer}}$ (1) $\Delta\theta_{\text{carpet}} = \frac{1225\ \textbf{ecf} \times 10\times10^{-3}}{0.045\times60} = 4.5$ and $\Delta\theta_{\text{underlay}} = 5.8$ and $\Delta\theta_{\text{concrete}} = 3.5$ (1) Add to give 13.8≈14 so claim is reasonable (1) Also accept use of $\frac{5\,000}{4}$ = 1250 W **Alternative:** Attempt at calculating $\frac{\Delta x}{KA}$ for 1 layer (1) $R_{\text{T(carpet)}} = \frac{\Delta x}{KA} = \frac{10\times10^{-3}}{0.045\times60} = 0.0037$ and $R_{\text{T(underlay)}} = 0.0048$ and $R_{\text{T(concrete)}} = 0.0029$ (1) Add to give 0.0114 and $\frac{\Delta Q}{\Delta t} = \frac{\Delta\theta}{\frac{\Delta x}{KA}} = 1228 \approx 1225$ so claim is reasonable (1)			3	3	3	